

Supplementary Material 4: Summary of Benchmark Algorithms Settings

Description: *This Supplementary material provides a summary of the main settings and configuration parameters of the algorithms employed for the comparative performance evaluation.*

1. A-LOAM

Repository: <https://github.com/HKUST-Aerial-Robotics/A-LOAM>

Table 1: Configuration parameters for A-LOAM.

Parameter	Value
Git commit	3f8b6d2
Lidar input Citrus and Pullally	/velodyne_points
Lidar input Bachuss	/os_cloud_node/points
Scan line	16
Skip mapping	10Hz (value =1)
Minimum range	1.2
Mapping line resolution	0.2
Mapping plane resolution	0.4

Change of the lidar input in file scanRegistration.cpp line:

For velodyne VLP-16:

```
ros::Subscriber subLaserCloud = nh.subscribe<sensor_msgs::PointCloud2> \
("/velodyne_points", 100, laserCloudHandler);
```

For ouster OS1-16:

```
ros::Subscriber subLaserCloud = nh.subscribe<sensor_msgs::PointCloud2> \
("/os_cloud_node/points", 100, laserCloudHandler);
```

2. LeGO-LOAM

Repository: <https://github.com/RobustFieldAutonomyLab/LeGO-LOAM>

Table 2: Configuration parameters for LeGO-LOAM.

Parameter	Value
Git commit	896a7a9
Lidar input Citrus and Pullally	/velodyne_points
Lidar input Bachuss	/os_cloud_node/points
Scan line	16
Maximum range	50.0
EdgeFeatureNum	2
SurfFeatureNum	4
Configuration file	include/utility.h
Loop closure	disabled

Change of the lidar input in file include/utility.h line:

For velodyne VLP-16:

```
extern const string pointCloudTopic = "/velodyne_points";
```

For Ouster OS1-16:

```
extern const string pointCloudTopic = "/os_cloud_node/points";
```

Since LeGO-LOAM requires the ring field in the point cloud, a preprocessing ROS node was used to inject this field for compatibility with the Ouster sensor data. in file include/utility.h line:

For velodyne VLP-16:

```
extern const bool useCloudRing = true;
```

For Ouster OS1-16:

```
extern const bool useCloudRing = false;
```

3. Direct LiDAR Odometry (DLO)

Repository: https://github.com/vectr-ucla/direct_lidar_odometry

Table 3: Configuration parameters for DLO.

Parameter	Value
Git commit	11528c0
Lidar input Citrus and Pullally	/velodyne_points
Lidar input Bachuss	/os_cloud_node/points
Configuration file	config/dlo.yaml

Change of the lidar input in launch file:

```
roslaunch direct_lidar_odometry dlo.launch pointcloud_topic:= /velodyne_points
```

For Ouster OS1-16:

```
roslaunch direct_lidar_odometry dlo.launch pointcloud_topic:= /os_cloud_node/points
```

4. MOLA

Repository: https://github.com/MOLAorg/mola_lidar_odometry

Table 4: Configuration parameters for MOLA.

Parameter	Value
Git version	1.3.1
Lidar input Citrus and Pullally	/velodyne_points
Lidar input Bachuss	/os_cloud_node/points
Configuration file	mola-cli-launches/lidar_odometry_ros2.yaml

It is possible to set the inputs in launch file:

For velodyne VLP-16:

```
ros2 launch mola_lidar_odometry ros2-lidar-odometry.launch.py \
lidar_topic_name:=/velodyne\_points \
ignore_lidar_pose_from_tf:=True \
publish_localization_following_rep105:=False
```

For Ouster OS1-16:

```
ros2 launch mola_lidar_odometry ros2-lidar-odometry.launch.py \
lidar_topic_name:=/os_cloud_node/points \
ignore_lidar_pose_from_tf:=True \
publish_localization_following_rep105:=False
```

The .bag files of all the datasets are not directly compatible with ROS 2. Therefore, it was necessary to convert them to the .mcap format. The conversion was performed using the following command:

```
rosbags-convert --src DATA_TO_BE_CONVERT.bag \
--dst CONVERTED_DATA.mcap --dst-storage mcap
```

5. AG-LOAM

Repository: <https://github.com/UCR-Robotics/AG-LOAM>

Table 5: Configuration parameters for AG-LOAM.

Parameter	Value
Git commit	9299d28
Lidar input Citrus and Pullally	/velodyne_points
Lidar input Bachuss	/os_cloud_node/points
Configuration file	config/params.yaml

Default parameters provided by the repository were used. To set the lidar input, modify the file inputconfig/params.yaml:

For velodyne VLP-16:

```
lidar\_topic: "/velodyne_points"
```

For Ouster OS1-16:

```
lidar\_topic: "/os_cloud_node/points"
```